

Security analysis and numerical simulation

1. Gate minimization using exclusive-or-sum-of-products (ESOP)

Exclusive-or sum-of-products (ESOP) is a two-level Boolean representation in which product terms are combined using XOR operations. ESOP minimization is particularly important in quantum computing because it can reduce the number of multi-controlled NOT (n-CNOT) gates. When multiple n-CNOT gates share the same target qubit, their combined effect can be represented as an ESOP expression, with each control set corresponding to a product term. In the NEQR framework [2], the Espresso algorithm [3] is used to minimize sum-of-products (SOP) expressions and reduce the number of n-CNOT gates. However, the resulting SOP expression must be converted into an ESOP form before implementation, and a minimal SOP does not necessarily yield a minimal ESOP. Therefore, direct ESOP minimization is often more effective, as it produces optimized ESOP expressions without requiring additional conversion. A comparison of ESOP minimization and the Espresso algorithm is provided in [4].

Consider the unitary operation of a general n -CNOT gate acting on n control qubits and a single target qubit. Let this operator be denoted by V_ϕ and its corresponding ESOP term be U_ϕ . The control configuration is represented by an n -bit binary sequence ϕ containing don't-care entries (*). Each position in ϕ specifies whether the corresponding qubit acts as a positive control, a negative control, or has no influence on the target qubit. If ϕ contains m don't-care positions, replacing each (*) with either 0 or 1 generates a set Φ of 2^m binary strings.

$$\begin{aligned}\phi &= \phi^0 \phi^1 \dots \phi^{n-1}, \quad \phi^i \in \{0, 1, *\} \quad 0 \leq i < n. \\ \Phi &= \{s \in \{0, 1\}^n \mid (\phi^i = *) \vee (\phi^i = s^i), 0 \leq i < n\}.\end{aligned}\tag{1}$$

The operator V_ϕ and the product term U_ϕ can be described as

$$V_\phi = \sum_{\substack{P=0 \\ P \notin \Phi}}^{2^n-1} I \otimes |P\rangle\langle P| + \sum_{\substack{P=0 \\ P \in \Phi}}^{2^n-1} X \otimes |P\rangle\langle P|,\tag{2}$$

$$U_\phi = \prod_{i=0}^{n-1} \nu_i, \quad \text{where } \nu_i = \begin{cases} v_i, & \phi_i = 1, \\ \bar{v}_i, & \phi_i = 0, \\ 1, & \phi_i = *. \end{cases} \quad (3)$$

The minimization procedure is as follows. Let us assume we have an operation composed a sequence of q multi-control COT gates, $F = \prod_{i=0}^{q-1} V_{\phi_i}$, operating on the same set of qubits. This corresponds to ESOP function $f = \bigoplus_{i=0}^{q-1} U_{\phi_i}$. Using ESOP minimization algorithms, the function can be minimized to get an equivalent form with t product terms, $f = \bigoplus_{i=0}^{t-1} U_{\phi_i}$. This expression can be used to construct the minimized quantum operation $F = \prod_{i=0}^{q-1} V_{\psi_i}$.

2. Illustrative quantum simulation of the proposed algorithm for a 4×4 matrix

A demonstration of the proposed algorithm on a quantum simulator utilizing a 4×4 quantum image is provided for verification and validation. The quantum circuit is simulated with the StatevectorSimulator provided by Qiskit Aer on a device featuring an Intel $i9\ 11^{th} Gen$, 16 GB RAM. To enhance clarity, only one iteration of the encryption procedure is illustrated, use a single S-box for the complete matrix. Additionally, the $m - CNOT$ gates are optimized via the EXORCISM-4 [5, 6] algorithm at every level. A demonstration of the proposed algorithm on a quantum simulator utilizing a 4×4 quantum image is provided for verification and validation. Table 1 presents a comparison of the circuit depths obtained with and without the implementation of EXORCISM-4 optimization. Similarly, Table 2 summarizes the quantum resource metrics associated with different unitary operations of the proposed algorithm for a 4×4 image.

Table 1: Comparison of quantum resource requirements without and with ESOP optimization.

Algorithm Component	CNOT Without ESOP	CNOT With ESOP	CNOT Reduction	T-depth Without ESOP	T-depth With ESOP	Circuit Depth Without ESOP	Circuit Depth With ESOP
GNEQR state preparation	1584	155	90.2%	1570	131	3850	341
Baker permutation	1677	248	85.2%	1570	131	3863	359
Chaotic diffusion	5680	1170	79.4%	5349	897	13442	2290
Quantum P-box	2748	427	84.5%	2718	335	6660	848
Complete encryption	7705	1649	78.6%	7141	1179	17992	3025
Encryption + inverse recovery	13826	3143	77.3%	12713	2225	32139	5704

Table 2: Quantum resource metrics for different components of the proposed encryption algorithm.

Algorithm Component	CNOT	T-depth	Ancilla	Circuit Width	Circuit Depth
GNEQR state preparation	155	131	8	20	341
Baker permutation	248	131	8	20	359
Chaotic diffusion	1170	897	8	20	2290
Quantum P-box	427	335	8	20	848
Complete encryption	1649	1179	8	20	3025
Encryption + inverse recovery	3143	2225	8	20	5704

2.1. GNEQR preparation

The plain image C , given in Eq.(4), is encoded in GNEQR to create the quantum plain image. The location information is stored in the registers y and x , and the pixel intensity is stored in the register F . Figure 1 depicts the quantum circuit of the 4×4 image.

$$\begin{pmatrix} 0 & 15 & 31 & 47 \\ 63 & 79 & 95 & 111 \\ 127 & 143 & 159 & 175 \\ 191 & 207 & 223 & 239 \end{pmatrix} \quad (4)$$

The expression of the quantum state $|C\rangle$ is as follows:

$$\begin{aligned} |C\rangle = \frac{1}{4}(&|00000000\rangle|0000\rangle + |00001111\rangle|0001\rangle + |00011111\rangle|0010\rangle + |00101111\rangle|0011\rangle \\ &+ |00111111\rangle|0100\rangle + |01001111\rangle|0101\rangle + |01011111\rangle|0110\rangle + |01101111\rangle|0111\rangle \\ &+ |01111111\rangle|1000\rangle + |10001111\rangle|1001\rangle + |10011111\rangle|1010\rangle + |10101111\rangle|1011\rangle \\ &+ |10111111\rangle|1100\rangle + |11001111\rangle|1101\rangle + |11011111\rangle|1110\rangle + |11101111\rangle|1111\rangle) \end{aligned} \quad (5)$$

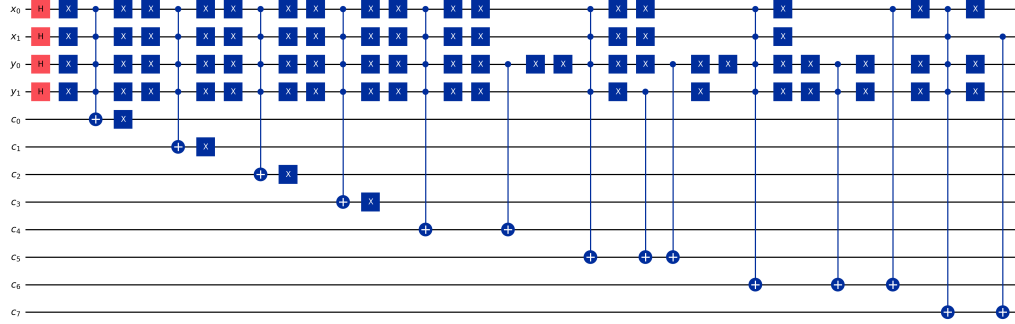


Figure 1: GNEQR quantum circuit.

2.2. Dual quantum Baker scrambling operations

After applying the dual quantum Baker scrambling on the state $|C\rangle$, resulting quantum state can be described as follows:

$$\begin{aligned}
 |I_1\rangle = \frac{1}{4} & (|00000000\rangle|0000\rangle + |11111101\rangle|0001\rangle + |00110100\rangle|0010\rangle + |11111110\rangle|0011\rangle \\
 & + |11110000\rangle|0100\rangle + |11110010\rangle|0101\rangle + |11110001\rangle|0110\rangle + |11110011\rangle|0111\rangle \\
 & + |11110100\rangle|1000\rangle + |11110110\rangle|1001\rangle + |11110101\rangle|1010\rangle + |11110111\rangle|1011\rangle \\
 & + |11111000\rangle|1100\rangle + |11111010\rangle|1101\rangle + |11111001\rangle|1110\rangle + |11111011\rangle|1111\rangle)
 \end{aligned} \tag{6}$$

Fig. 2 illustrates the proposed dual quantum Baker permutation applied to the position and grayscale-value registers of the GNEQR-encoded image.

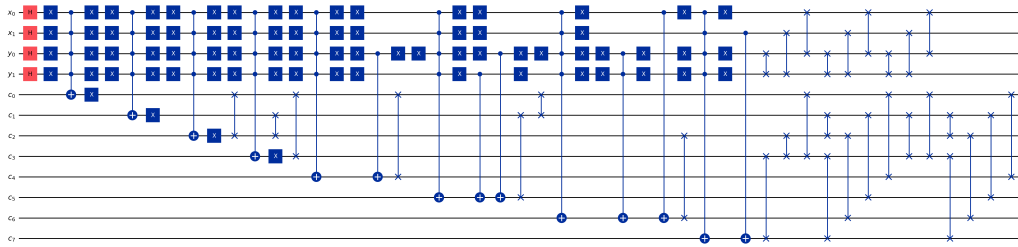


Figure 2: Dual Baker quantum circuit.

2.3. Quantum chaotic XOR diffusion

After applying the quantum operator U the state $|I_1\rangle$ can be illustrated as follows:

$$\begin{aligned}
 |I_2\rangle = \frac{1}{4}(&|01110001\rangle|0000\rangle + |11110101\rangle|0001\rangle + |10100100\rangle|0010\rangle + |00110110\rangle|0011\rangle \\
 &+ |11101011\rangle|0100\rangle + |00110110\rangle|0101\rangle + |10011011\rangle|0110\rangle + |10001100\rangle|0111\rangle \\
 &+ |01010011\rangle|1000\rangle + |00101000\rangle|1001\rangle + |10011101\rangle|1010\rangle + |10000101\rangle|1011\rangle \\
 &+ |01110111\rangle|1100\rangle + |11000101\rangle|1101\rangle + |11011001\rangle|1110\rangle + |01000010\rangle|1111\rangle)
 \end{aligned}
 \tag{7}$$

Fig. 3 depicts the quantum circuit diagram of XOR diffusion.

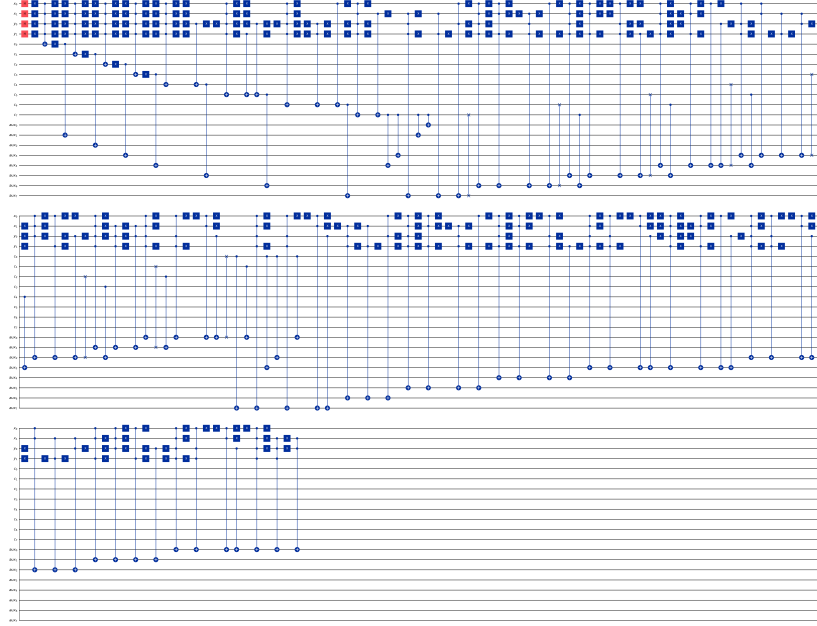


Figure 3: Chaotic XOR quantum circuit.

2.4. Permutation box operator

After applying the quantum unitary operator Γ on quantum state $|I_2\rangle$, resulting quantum state can be expressed as follows:

$$\begin{aligned}
 |I_3\rangle = \frac{1}{4}(&|01010011\rangle|0000\rangle + |00101000\rangle|0001\rangle + |10011011\rangle|0010\rangle + |11110101\rangle|0011\rangle \\
 &+ |10011101\rangle|0100\rangle + |00110110\rangle|0101\rangle + |11000101\rangle|0110\rangle + |10000101\rangle|0111\rangle \\
 &+ |11101011\rangle|1000\rangle + |01110001\rangle|1001\rangle + |10001100\rangle|1010\rangle + |01110111\rangle|1011\rangle \\
 &+ |11011001\rangle|1100\rangle + |00110110\rangle|1101\rangle + |10100100\rangle|1110\rangle + |01000010\rangle|1111\rangle)
 \end{aligned}
 \tag{8}$$

Fig. 4 illustrates the proposed quantum P-box permutation applied to the position register of the diffused quantum state $|I_2\rangle$.

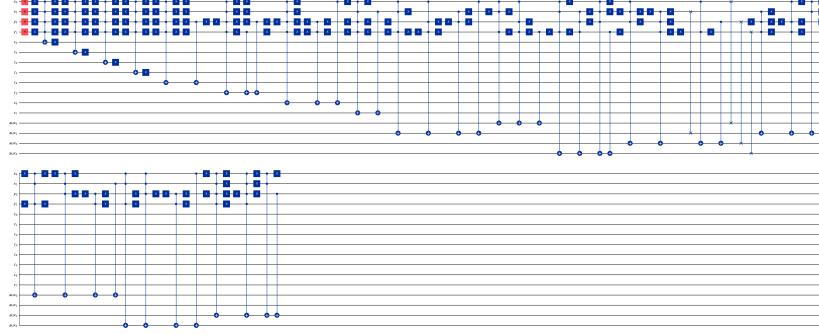


Figure 4: Proposed P-box quantum circuit.

3. Security and numerical analysis

3.1. Visual and histogram analysis

This subsection computes the encryption and concealment performance of the proposed algorithm through histogram analysis, as shown in Fig. 5. The histogram of the plain image exhibits noticeable pixel distribution patterns, whereas the histogram of the encrypted image demonstrates a nearly uniform distribution [7], indicating strong diffusion characteristics and effective resistance against statistical attacks.

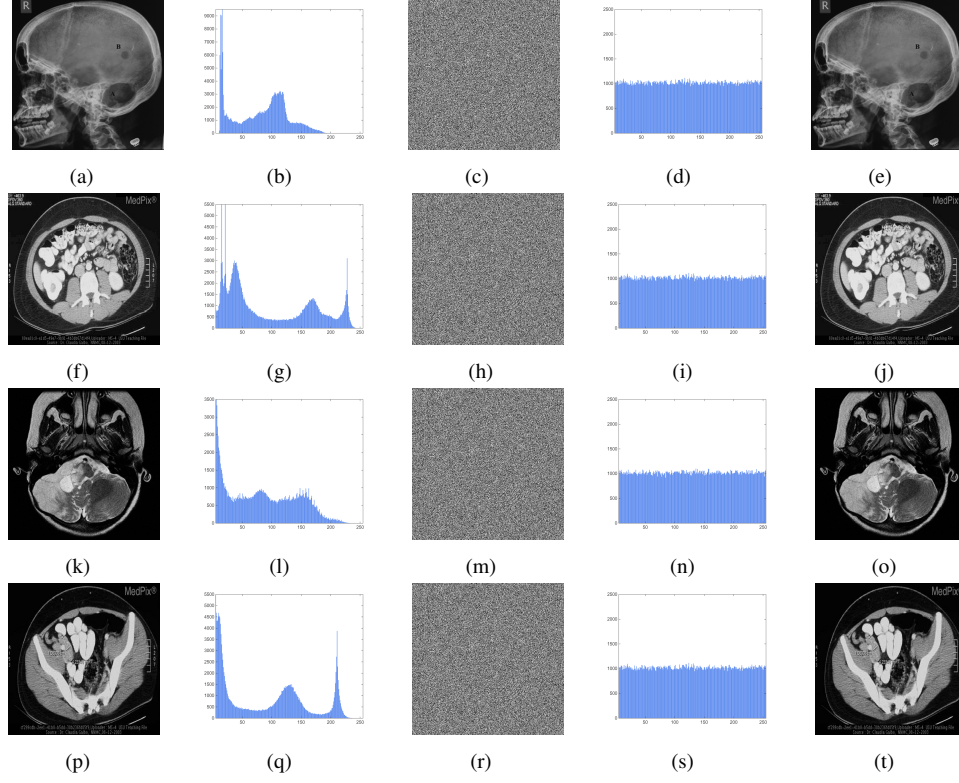


Figure 5: Histogram analysis: rows depict medical images (Brain, Cecum) and their corresponding histograms, encrypted images and their corresponding histograms, reconstructed images.

3.2. Cryptanalytic attack analysis

Cryptanalytic attacks attempt to exploit structural relations among plaintexts, ciphertexts, and encryption parameters to recover the secret key or an equivalent decryption transformation. To evaluate the cryptanalytic robustness of the proposed quantum-inspired image encryption scheme, chosen-plaintext, known-plaintext, and divide-and-conquer attacks are considered. Let

$$C = \text{Enc}(P; K_{\text{img}}, N) \quad (9)$$

denote the encryption of a plaintext image P , where $K_{\text{img}} \in \{0, 1\}^{256}$ is the secret image key and $N \in \{0, 1\}^{96}$ is a public nonce generated independently for each encryption instance. The plaintext-binding hash h , system parameters, attractor bounds, and the complete encryption procedure are assumed to be known to the adversary, whereas K_{img} remains secret.

As defined in Algorithm 2, the plaintext-binding hash is computed as

$$h = \text{SHA3-256}(P \parallel \text{width}(P) \parallel \text{height}(P) \parallel \text{channels}(P)), \quad (10)$$

and the pseudorandom key is obtained as

$$\text{PRK} = \text{HKDF-Extract}_{\text{SHA256}}(N, K_{\text{img}} \parallel h). \quad (11)$$

Thus, the pair (N, h) provides encryption-instance diversification: h binds the derived control information to the plaintext, whereas the fresh nonce N distinguishes independent encryption instances, including repeated encryptions of the same image.

The domain-separated key-derivation procedure subsequently generates

$$\mathbf{x}^{(b)}(0), \quad \mathbf{x}^{(d)}(0), \quad \mathbf{x}^{(p)}(0), \quad (12)$$

which control the dual Baker permutation (DBP), diffusion, and P-box stages, respectively. In particular, $\mathbf{x}^{(b)}(0)$ generates the round-selector sequences $\{k_t\}_{t=1}^R$ and $\{k'_t\}_{t=1}^R$, $\mathbf{x}^{(d)}(0)$ generates the diffusion sequence D , and $\mathbf{x}^{(p)}(0)$ generates the bijective P-box P_b .

Let

$$J = \text{DBP}_{\{k_t\}, \{k'_t\}}^{(R)}(P) \quad (13)$$

denote the output of the R -round dual Baker permutation. The diffusion stage is represented by

$$J' = \text{Diff}_D(J), \quad (14)$$

where Diff_D denotes the XOR-based quantum diffusion defined in Sect. 3.5.3, including the binary intensity transformation and the chaotic sequence D . The final P-box operation gives

$$C = \Gamma_{P_b}(J'). \quad (15)$$

Hence, the overall encryption transformation can be expressed as

$$C = \Gamma_{P_b} \left(\text{Diff}_D \left[\text{DBP}_{\{k_t\}, \{k'_t\}}^{(R)}(P) \right] \right). \quad (16)$$

3.2.1. Resistance to chosen-plaintext attacks

In a chosen-plaintext attack (CPA), an adversary selects specially structured plaintexts and observes their corresponding ciphertexts in an attempt to identify reusable permutation or diffusion patterns. Constant-intensity, row-structured,

column-structured, and checkerboard images are particularly useful for detecting residual spatial or intensity relationships.

For this analysis, the following structured plaintext set is considered:

$$\mathcal{P}_{\text{CPA}} = \{P_{\text{black}}, P_{\text{white}}, P_{\text{gray}}, P_{\text{row}}, P_{\text{column}}, P_{\text{checker}}\}. \quad (17)$$

To provide a conservative attack setting, the same secret K_{img} may be assumed across the adversary's encryption queries. However, each encryption instance uses a fresh nonce N_i , as prescribed by Algorithm 2, while its plaintext-dependent hash is

$$h_i = H(P_i). \quad (18)$$

Therefore, for the i th query,

$$\text{PRK}_i = \text{HKDF-Extract}_{\text{SHA256}}(N_i, K_{\text{img}} \| h_i). \quad (19)$$

For two distinct chosen plaintexts $P_i \neq P_j$, the corresponding hashes h_i and h_j differ except with negligible collision probability. In addition, independently generated nonces N_i and N_j differ with overwhelming probability. Since both quantities participate in the key derivation, different encryption instances generate distinct stage-specific chaotic states and, consequently, different DBP selectors, diffusion sequences, and P-box permutations.

The nonce also provides diversification when the same plaintext is encrypted more than once. If $P_i = P_j$, then $h_i = h_j$; nevertheless, fresh nonces $N_i \neq N_j$ produce different PRKs and hence different encryption-control sequences. Therefore, the internal permutation and diffusion information associated with one chosen-plaintext query is not directly reusable in another encryption instance. To evaluate the statistical resilience of the scheme under a CPA scenario, different images of size 512×512 were encrypted. The resulting cipher images exhibit near uniform histograms as shown in Fig. 6 along side the statistical properties of the ciphertexts obtained from the structured plaintexts are summarized in Table 3.

The resulting ciphertexts exhibit entropy values between 7.9992 and 7.9993, which are close to the ideal value of 8 for an 8-bit image. Furthermore, the horizontal, vertical, and diagonal correlation coefficients remain close to zero despite the highly structured nature of the selected plaintexts. These results indicate that the combined DBP, diffusion, and P-box transformations effectively suppress exploitable plaintext structures, supporting the resistance of the proposed scheme to chosen-plaintext attacks.

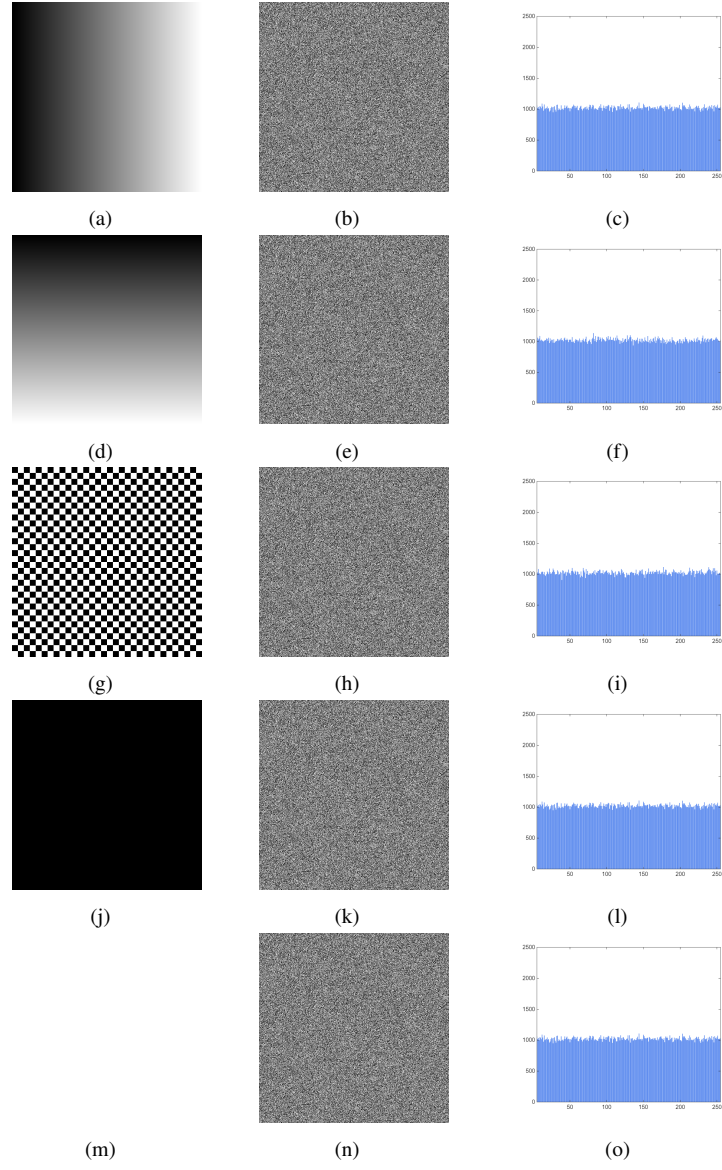


Figure 6: Histogram analysis: first column depict different images, second column present corresponding encrypted images and third column illustrate the corresponding histograms.

Table 3: Statistical metrics of ciphertexts obtained from structured chosen plaintexts.

Image	Entropy	Correlation coefficient		
		H	V	D
P_{black}	7.9992	-0.0024	-0.0056	0.0012
P_{checker}	7.9993	-0.0014	-0.0118	-0.0026
P_{white}	7.9993	-0.0042	-0.0131	-0.0115
P_{stripe}	7.9993	-0.0065	-0.0088	-0.0031
P_{row}	7.9993	-0.0178	-0.0075	0.0010
P_{column}	7.9993	-0.0092	-0.0105	-0.0061
P_{gray}	7.9993	-0.0385	-0.0203	-0.0002

Differential chosen-plaintext analysis. Differential chosen-plaintext analysis evaluates the sensitivity of the complete encryption transformation to a minimal plaintext modification. Let P_1 and P_2 differ by only one bit of a single pixel. In this experiment, both the secret key and the nonce are deliberately kept identical:

$$C_1 = \text{Enc}(P_1; K_{\text{img}}, N), \quad C_2 = \text{Enc}(P_2; K_{\text{img}}, N). \quad (20)$$

Keeping N fixed is important in this experiment because it isolates the effect of the plaintext modification rather than introducing additional variation from a new nonce. The one-bit change nevertheless produces

$$h_1 = H(P_1) \neq H(P_2) = h_2, \quad (21)$$

except with negligible collision probability. Consequently, the change in h propagates through the complete key-derivation mechanism and modifies the DBP selectors, diffusion sequence, and P-box permutation. The experiment therefore measures the plaintext-avalanche behavior of the complete proposed construction.

For two cipher images of size $H \times W$, the number of pixels change rate (NPCR) is defined as

$$\text{NPCR} = \frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W D(i, j) \times 100\%, \quad (22)$$

where

$$D(i, j) = \begin{cases} 0, & C_1(i, j) = C_2(i, j), \\ 1, & C_1(i, j) \neq C_2(i, j). \end{cases} \quad (23)$$

The unified average changing intensity (UACI) is given by

$$\text{UACI} = \frac{1}{255HW} \sum_{i=1}^H \sum_{j=1}^W |C_1(i, j) - C_2(i, j)| \times 100\%. \quad (24)$$

For independent uniformly distributed 8-bit cipher pixels, the corresponding reference values are approximately

$$\text{NPCR}_{\text{ideal}} = 99.6094\%, \quad \text{UACI}_{\text{ideal}} = 33.4635\%. \quad (25)$$

The proposed scheme obtains

$$\text{NPCR} = 99.6043\%, \quad \text{UACI} = 33.4899\%. \quad (26)$$

These values are close to their theoretical references, demonstrating that a one-bit modification of the plaintext produces extensive changes throughout the ciphertext. This confirms a strong plaintext-avalanche response and supports resistance to differential chosen-plaintext attacks.

3.2.2. Resistance to known-plaintext attacks

In a known-plaintext attack (KPA), the adversary possesses one or more plaintext-ciphertext pairs (P_i, C_i) together with the associated public values (N_i, h_i) and attempts to recover K_{img} or an equivalent reusable decryption transformation. For a known pair (P, C) , the encryption relation is

$$C = \Gamma_{P_b} \left(\text{Diff}_D \left[\text{DBP}_{\{k_t\}, \{k'_t\}}^{(R)}(P) \right] \right). \quad (27)$$

Knowledge of the plaintext does not directly reveal the intermediate DBP output

$$J = \text{DBP}_{\{k_t\}, \{k'_t\}}^{(R)}(P), \quad (28)$$

because the DBP transformation depends on secret-derived round selectors.

Similarly, the input to the diffusion stage cannot be obtained directly from the ciphertext because the final coordinate arrangement depends on the unknown P-box P_b .

Using the diffusion formulation of Sect. 3.5.3, the chaotic diffusion mask can be isolated only after the correct DBP output and inverse P-box are known. Writing the known pixelwise linear part of the diffusion as $\mathcal{L}$, one obtains

$$D = \Gamma_{P_b}^{-1}(C) \oplus \mathcal{L} \left(\text{DBP}_{\{k_t\}, \{k'_t\}}^{(R)}(P) \right). \quad (29)$$

Thus, the known pair (P, C) alone does not expose the diffusion sequence, because both the DBP round selectors and the P-box permutation remain unknown. Moreover, each encryption instance is associated with the public pair (N_i, h_i) . A different plaintext changes h_i , whereas a fresh nonce changes N_i even when the same plaintext is encrypted repeatedly. Consequently, under a fixed K_{img} , different instances generate different Baker, diffusion, and P-box control sequences. Information obtained from one known plaintext-ciphertext pair therefore does not directly provide a reusable internal transformation for another instance. Under the actual per-image key-generation procedure of Algorithm 2, the independently generated master key provides an additional level of separation between encryption instances.

Hence, the combination of secret-dependent DBP scrambling, instance-specific diffusion and P-box generation, and (N, h) -based diversification supports resistance against the considered known-plaintext attack.

3.2.3. Resistance to divide-and-conquer attacks

A divide-and-conquer attack attempts to decompose a composite encryption scheme into smaller components that can be recovered independently. In the proposed construction, an adversary may attempt to determine the DBP round selectors, the chaotic diffusion sequence, and the P-box permutation separately. Their derivation can be summarized as

$$(K_{\text{img}}, N, h) \longrightarrow \begin{cases} \mathbf{x}^{(b)}(0) \longrightarrow \{k_t\}_{t=1}^R, \{k'_t\}_{t=1}^R, \\ \mathbf{x}^{(d)}(0) \longrightarrow D, \\ \mathbf{x}^{(p)}(0) \longrightarrow P_b. \end{cases} \quad (30)$$

The three initial states are obtained through domain-separated key derivation. Hence, knowledge of the output of one branch does not directly determine the other two branches. At the same time, the resulting transformations remain coupled through

$$C = \Gamma_{P_b} \left(\text{Diff}_D \left[\text{DBP}_{\{k_t\}, \{k'_t\}}^{(R)}(P) \right] \right). \quad (31)$$

For example, inversion of a candidate P-box only exposes the output of the unknown diffusion stage. Similarly, knowledge of a candidate diffusion sequence is insufficient without the correct DBP-transformed image, while a candidate DBP control sequence remains obscured by the subsequent diffusion and P-box transformations. Thus, recovering one stage independently does not directly yield the complete decryption transformation.

The DBP round selectors $\{k_t\}$ and $\{k'_t\}$ are generated from the single Baker-stage chaotic trajectory initialized by $\mathbf{x}^{(b)}(0)$ and therefore constitute one derived control sequence rather than independent secret keys for each round. In addition, the fresh pair (N, h) changes the derived control material from one encryption instance to another, preventing an instance-specific intermediate component from being directly reused across different encryptions.

Finally, the security root of the key-generation procedure is the 256-bit secret K_{img} . As established in the key-space analysis, an exhaustive classical search requires 2^{256} candidate keys, whereas an idealized Grover search requires on the order of 2^{128} evaluations. Together with domain-separated key derivation and the coupling of the DBP, diffusion, and P-box transformations, this supports the resistance of the proposed scheme against the considered divide-and-conquer attack.

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